AKENTEN APPIAH-MENKA UNIVERSITY of Skills Training and Entrepreneurial Development

FACULTY OF APPLIED SCIENCES AND MATHEMATICS EDUCATION
DEPARTMENT OF INFORMATION TECHNOLOGY EDUCATION

COURSE OUTLINE FOR FIRST SEMESTER OF 2025/2026 ACADEMIC YEAR

COURSE CODE/TITLE: ICT233; Web Technologies and Design. **CREDIT HOURS**: 3

PROGRAMME: BSc. Information Technology Education **CLASS:** Level 200

COURSE LECTURER: Maxwell Dorgbefu Jnr. **OFFICE**: FBR 31, FTE Block

PHONE: +233 (0) 243 337 486

COURSE INTRODUCTION

The need to be present on the World Wide Web (WWW) by private business enterprises, government agencies, and individuals is crucial for doing business with prospective customers and staying in touch with friends and family. This need is accomplished by using web sites and web applications. These web-based platforms are planned, designed, and deployed using tools, that are collectively referred to as Web Technologies. This course is aimed at providing students with knowledge, skills, and relevant competencies in the key concepts, software, and technologies for planning, designing, and deploying robust web applications.

COURSE CONTENT

The following topics constitute the course content:

Introduction – Fundamentals of web technologies and web programming, The World Wide Web, Web servers and Web browsers, Domain name registration and web hosting, the web design process, Overview of HTML, HTML Document Structure, Creating web pages using HTML, CSS rules, CSS Selectors, The CSS Box Model, Layout design with CSS, Overview of JavaScript, variables and data types, JS Syntax, objects and arrays, control flow and functions in JS, Document Object Model (DOM) manipulation and events, and asynchronous JS.

COURSE OBJECTIVES

The aim of the course is to provide students with knowledge, skills, and related competencies required for planning, designing, developing, and deploying state-of-the-art web applications that support the 21st century enterprises.

The specific objectives will be dealt with on a lecture-by-lecture basis.

By the end of the course the student should be able to:

1. Explain the fundamental concepts of web development
2. Create web pages using HTML.
3. Style web pages with CSS in a professional manner
4. Use JavaScript in web development.
5. Develop real-world web sites and UI of web applications

OUTLINE OF INSTRUCTION

To achieve the set objectives of this course, the course is organized into the following thematic areas:

Theme I: Overview of Web Technologies and Web Design Processes

Theme II: Creating Web Contents using Hypertext Markup Language (HTML)

Theme III: Designing Presentation layer of web applications using Cascading Style Sheets (CSS).

Theme IV: JavaScript in Web Development.

MODE OF DELIVERY AND ASSESSMENT

The course shall be delivered using lectures (face-to-face, and online), and practical computer lab sessions. The tentative assessment components of the course are outlined in the table below:

S/N	ACTIVITY	REWARD (%)
1.	Pop Quiz and Mid-Semester Quiz	15
2.	Lab Sessions and Semester's Project	20
3.	Class Attendance and Participation	5
4.	End of semester examination	60
	Total	100

TENTATIVE WEEKLY LECTURE SCHEDULES

Week 1: Overview of Web Technologies

- Introduction to the course
- Definition and explanation of key terminologies.
- A brief introduction to some popular web technologies.

Week 2: The web design process

- Web browsers and web servers
- The key steps involved in developing web applications.
- Domain Registration and Hosting

Weeks 3 - 7: Introduction to HTML, and CSS

- HTML Elements and Tags, and the HTML Document Structure
- Creating HTML Documents and Previewing in a Browser
- Creating Lists and Links in HTML
- Adding Multimedia to Web pages
- Presenting Information on Web pages using HTML Tables
- Accepting user inputs with HTML Forms
- HTML5 Features and APIs
- Fundamentals of CSS and The CSS Box Model
- Layout design in CSS

Week 8 – 11: Introducing Interactivity into Web Applications using JavaScript

- Fundamentals of JavaScript
- Arrays and objects in JavaScript
- Control Flow and Functions in JavaScript
- DOM Manipulation and events in JavaScript
- Asynchronous JavaScript
- Error Handling and Debugging Techniques

Weeks -12: Project Presentation and Course Review.

Weeks 13-16: End of Semester Examinations.

GRADING POLICY

The grading policy shown below will be used in the determination of your final grades in this course.

S/N	Score (%)	Grade	Grade Point
1.	80 – 100	A	4.0
2.	75 – 79	B+	3.5
3.	70 – 74	B	3.0
4.	65 – 69	C+	2.5
5.	60 – 64	C	2.0
6.	55 – 59	D+	1.5
7.	50 – 54	D	1.0
8.	0 – 49	E	0

HONOR CODE

The Code of Honor will be strictly applied as described in the **Handbook on Examinations and Review of Academic Programmes of AAMUSTED**. Students will not give or receive aid on quizzes, class exercises, and end of semester examinations. This includes, but is not limited to, viewing the works or submissions of others, sharing answers with others, and making unauthorized use of books or notes, and electronic gadgets while taking these assessments. On group projects, teams are encouraged and must work independently. Note that, relying on solutions from other groups, whether they are currently on the course, constitutes plagiarism. On AI, use it responsibly as we await a comprehensive policy.

SUGGESTED REFERENCES/READING LIST

1. Connoly, R. & Hoar, R. (2022). Fundamentals of Web Development (3rd ed.). Pearson
2. Bramer, M. (2016). Web Programming with PHP and MySQL. Springer
3. Carey, P. (2018). New Perspectives of HTML5, CSS3, and JavaScript (6th ed.). Cengage Learning
4. McGrath, M. (2020). HTML, CSS & JavaScript in Easy Steps (Special ed.). In Easy Steps.
5. Prettyman, S. (2020). Learn PHP 8 Using MySQL, JavaScript, CSS3, and HTML5 (2nd ed.). Apress
6. Robbins, J.N. (2018). Learning Web Design Beginner's Guide to HTML, CSS, JavaScript, and Web Graphics (5th ed.). O'Reilly Media Inc.
7. https://www.w3schools.com
8. https://developer.mozilla.org/en-US/docs/Web/Tutorials